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CONSTRUCTION ANALYTICS



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CLIENT FEEDBACK | DATA VISUALIZATION | GEOSPATIAL ANALYTICS

CONSTRUCTION ANALYTICS

PRODUCT CASE STUDY

A full-stack feedback platform that transforms construction project preferences and locations into a clear analytics experience through a validated form, dashboard, interactive map, and CSV export.

ROLE	PLATFORM	STACK
Product / UX-UI Designer + Front-End Developer	Responsive Web App	Next.js 14 TypeScript Tailwind CSS
Focus: Feedback → Insights		React Query Recharts Leaflet OpenStreetMap

LIVE DEMO → <https://construction-feedback-app.vercel.app/>
SOURCE CODE → <https://github.com/mahsakhakpour/construction-feedback>

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DOCUMENTATION NOTE

The supplied project description documents the implemented features and technology. It does not include recorded interviews, survey results, validated personas, or a documented sequence of user-tested iterations. Those sections are therefore framed as product-design artifacts and validation plans derived from the implemented experience, not as invented research results.

01 - EXECUTIVE SUMMARY

From Feedback to Actionable Construction Insights

Construction Analytics is a full-stack web platform for collecting structured client feedback about construction project types and locations, then making that information useful through visual analytics and geographic context.

The product follows a simple information lifecycle: collect the response, validate it, convert location into geographic coordinates, visualize the dataset, explore individual locations, and export the underlying records.

CORE PRODUCT IDEA

Transform a simple client feedback form into a visual, geographic, and actionable source of insight.

INPUT	ANALYSIS
Company name, project location, project type	Totals, category distribution, quick statistics

GEOSPATIAL	EXPORT
Interactive map with project-type markers	CSV download for external analysis

02 - PRODUCT CONTEXT & PROBLEM

The Problem

A construction feedback form can collect valuable information, but raw responses become harder to interpret when they remain as disconnected submissions. The product opportunity is to turn those records into a coherent decision-support experience.

DESIGN CHALLENGE

How might we turn simple construction client feedback into a clear, visual, geographic, and reusable source of insight?

Problem Areas

- Feedback collection and analysis are often separated.
- Category patterns are difficult to see in raw records.
- Location information has additional value when viewed geographically.
- Users may need the underlying data outside the application.

Opportunity

Create one connected workflow in which the submission form is not the end of the experience, but the first step in a pipeline from input to insight.

03 - GOALS & SUCCESS CRITERIA

Product Goals

GOAL	WHAT IT MEANS
Reduce friction	Keep the client feedback form focused and understandable.
Make patterns visible	Surface totals and project-type distribution immediately.
Add geographic context	Show where responses are located and distinguish project types.
Support external workflows	Allow the dataset to be exported as CSV.
Connect input and output	Ensure submitted responses become visible in analytics without a manual workflow.

Success Criteria

- A first-time respondent can understand what information is required.
- An analytics user can identify the most popular project type quickly.
- A user can locate and inspect an individual project geographically.
- A user can export the response dataset for additional analysis.

04 - RESEARCH & DISCOVERY

Research Approach

The supplied project description does not include recorded user interviews or survey results. Rather than invent research findings, this case study uses a lightweight product-discovery framework based on the actual implemented features.

SOURCE	WHAT IT TELLS US
Existing product scope	The platform is designed around project preferences and locations.
Implemented UI	The dashboard prioritizes response volume, project-type distribution, and quick statistics.
Map interaction	Location is important enough to warrant a dedicated geographic experience.
CSV export	Users may need the underlying records for work outside the application.
Technology choices	The product requires form validation, data synchronization, visualization, geocoding, and mapping.

Research Questions for Validation

- Which project metrics do users need to see first?
- Do users understand the project-type colour coding without explanation?
- How useful is geographic clustering for construction decision-making?
- Which fields are essential in an individual project record?
- How frequently do users need to export data?
-

RESEARCH STATUS

These are validation questions, not claimed findings. They can become the basis for future interviews or usability testing.

05 - PROTO-PERSONAS

These proto personas are derived from the product's two main usage contexts. They are hypotheses until validated through interviews.

Proto-Persona 01 - Maya, Client Representative

ATTRIBUTE	HYPOTHESIS
Role	Construction client/ project representative
Goal	Submit project information quickly and accurately.
Pain points	Long forms, unclear required fields, repeated data entry.
Needs	Clear labels, fast validation, obvious submission state.
Success	Submission completes successfully, and the information is captured correctly.

Proto-Persona 02 - David, Project Manager

ATTRIBUTE	HYPOTHESIS
Role	Project/operations manager
Goal	Understand response patterns and locations.
Pain points	Raw spreadsheets, fragmented location information, manual counting.
Needs	At-a-glance metrics, map context, export.
Success	Can understand the dataset quickly and export it when needed.

06 - USER NEEDS & REQUIREMENTS

From User Needs to Product Requirements

USER NEED	PRODUCT RESPONSE	FEATURE
Submit information easily	Short structured form	Feedback Form
Know what is required	Real-time validation	React Hook Form
Understand overall volume	Top-level metric cards	Dashboard
Compare categories	Pie chart + progress bars	Dashboard
Understand location patterns	Interactive markers	Leaflet Map
Inspect individual records	Marker popup details	Map
Reuse collected data	One-click download	CSV Export

Functional Requirements

- Capture company/client name.
- Capture project location.
- Capture project type.
- Validate form input.
- Persist or retrieve response data.
- Calculate category counts and percentages.
- Geocode locations into coordinates.
- Render project locations on a map.
- Export response records as CSV.

07 - INFORMATION ARCHITECTURE

The product has a deliberately small navigation structure so users can move between collection and analysis without learning a complex information hierarchy.

01	HOME / FEEDBACK Primary entry point for submitting a construction project response.
02	SURVEYS / RESPONSES Navigation context for the feedback workflow.
03	DASHBOARD High-level analytics and project distribution.
04	MAP Geographic exploration of project locations.
05	EXPORT Download the underlying response data as a CSV.

NAVIGATION PRINCIPLE

Keep the highest-value destinations visible while allowing the dashboard to function as the main analysis hub.

08 - USER FLOW

End-to-End User Flow

The nine steps below trace the full path from opening the form to exporting the dataset.



Figure — End-to-end user flow, generated for this case study.

01	OPEN FORM User arrives at the feedback form.
02	ENTER DETAILS Company name and project location are entered.
03	SELECT TYPE Residential, Commercial, or Industrial is selected.
04	VALIDATE The form checks required information.
05	SUBMIT The response is saved and becomes part of the dataset.
06	GEOCODE The location is converted to coordinates.
07	ANALYZE Dashboard cards and charts update from the response data.
08	EXPLORE User can inspect project distribution geographically.
09	EXPORT User can download the dataset as CSV.

09 - MIND MAP



Figure — Radial mind map, generated for this case study.

BRANCH	DETAILS
Collect	Company • Location • Project Type
Validate	Required fields • Error handling
Analyze	Totals • Categories • Percentages
Map	Coordinates • Markers • Popups
Export	CSV • Spreadsheet workflows
Improve	Filters • Trends • Database • Accessibility

VISUALIZATION OPTION

This radial diagram was generated to match the dashboard's colour palette. Swap it for a Figma-exported version any time — keep Construction Analytics in the centre and the same branch colours.

10 - JOURNEY MAP

Client-to-Insight Journey Map

This journey map represents the intended end-to-end experience based on the implemented product flow. It is a product-design model, not a documented research finding.

	SUBMIT	VALIDATE	ANALYZE	EXPLORE	EXPORT
User Goal	Provide project details	Know data is valid	Understand patterns	Understand location	Reuse dataset
Action	Fill form + choose type	Correct errors + submit	Read cards/charts	Open map + markers	Download CSV
Thinking	"This should be quick."	"Did it save correctly?"	"What is most common?"	"Where are these projects?"	"I need this in my sheet."
Friction	Too many fields	Unclear errors	Information overload	Map clutter	Unclear export state
Opportunity	Simple form	Immediate feedback	Strong hierarchy	Consistent markers	Clear export CTA

11 - WIREFRAMES

These wireframe structures map directly to the implemented product. They document the layout logic rather than claiming that the original project used a particular low-fidelity iteration.

Wireframe 01 — Feedback Form

AREA	CONTENT
Header	Logo/navigation
Main	Page title + short instruction
Form	Company Name • Location • Project Type
Action	Submit + validation
Footer	Supporting information

Wireframe 02 — Dashboard

AREA	CONTENT
Header	Navigation + Export CTA
Overview	4 metric cards
Analysis	Project distribution chart
Quick Stats	Most popular • Total responses • Categories
Detail	Optional deeper analysis

Wireframe 03 — Map

AREA	CONTENT
Header	Navigation
Map	Full-width geographic canvas
Controls	Zoom / fit locations
Markers	Project-type color coding
Popup	Client • Type • Address

Wireframe 04 — CSV Export

AREA	CONTENT
Dashboard	Export to CSV CTA
Action	Generate / download CSV
Data	Client • Type • Address • Timestamp
External	Excel / Sheets/analysis tools

12 - DESIGN SYSTEM & VISUAL LANGUAGE

Visual Language

The supplied dashboard establishes a dark analytics interface with blue, green, and yellow category accents. The case study uses the same visual logic for diagrams and documentation.

ELEMENT	USE
Dark navy	Primary navigation and analytics canvas
Blue	Residential/primary interactive accent
Green	Commercial / success / positive data state
Yellow	Industrial / attention/category accent
Purple	Secondary analytics callout
Light neutral	Cards, wireframes, and documentation backgrounds

Dashboard-Aligned Design Rules

- Use the same project-type colours consistently across cards, chart labels, progress bars, and map markers.
- Use high-contrast white text on the dark dashboard surface.
- Reserve accent colours for meaning and actions rather than decoration.
- Use compact cards and clear section headers for data-heavy screens.
- Keep analytics visually scannable before requiring deeper interaction.

13 - FEEDBACK FORM

Home Page — Feedback Form

The home page is the primary collection experience. Clients enter company name, project location, and project type, with React Hook Form used for validation and error handling.

The screenshot shows a web application titled "Construction Analytics". It features a navigation bar with links for Home, Surveys, and Dashboard. The main content is divided into two sections: "Submit Your Feedback" and "Feedback Results".

Submit Your Feedback

Client Name
e.g., ABC Construction

Project Type

- ☒ Residential Construction
- ☐ Commercial Construction
- ☐ Industrial Construction

Submit Feedback

Feedback Results

A pie chart displays the distribution of feedback by project type:

- Commercial: 33%
- Residential: 22%
- Industrial: 44%

Below the chart, a legend identifies the colors: Commercial (green), Industrial (yellow), and Residential (blue). At the bottom, three large numbers represent the count for each type: 2 Residential, 3 Commercial, and 4 Industrial.

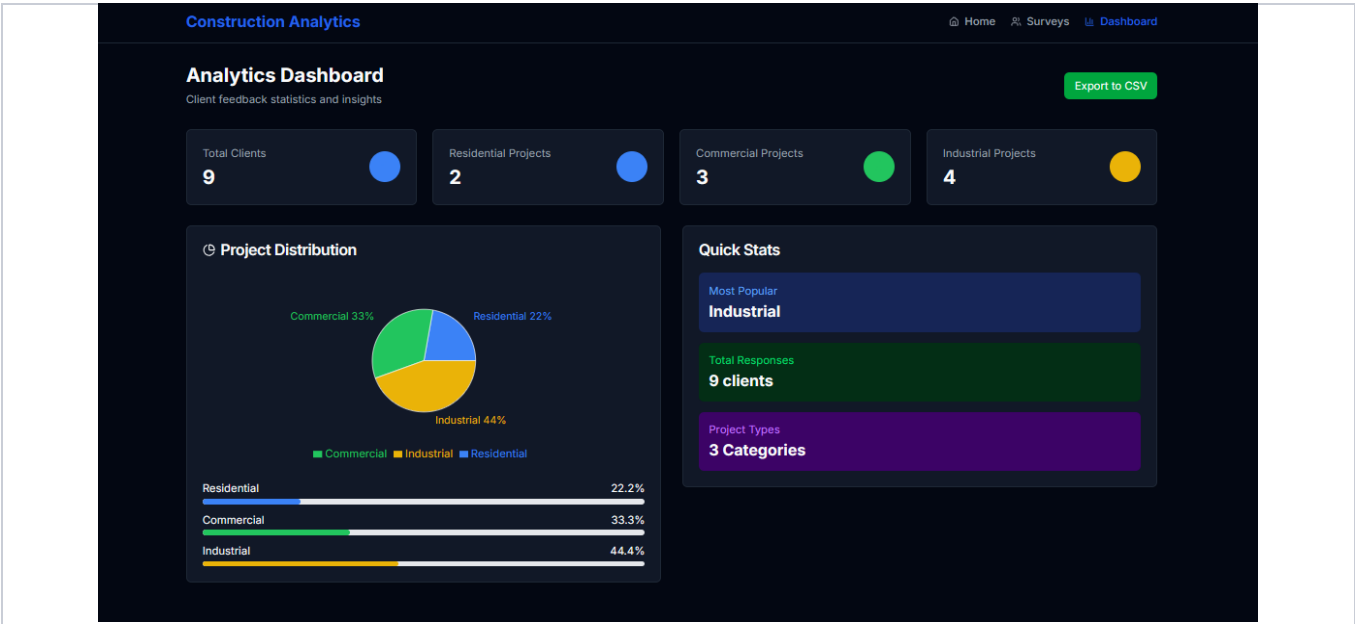
Home Page — Feedback Form (home.png, supplied).

UX Focus

- Clear field labels
- Minimal required information
- Immediate validation
- Obvious submit action
- Confirmation after successful submission

14 - ANALYTICS DASHBOARD

The dashboard is the central analysis surface. It combines high-level metrics with category distribution and quick statistics so a user can understand the dataset before exploring individual locations.



Analytics Dashboard (dashboard.png, supplied).

COMPONENT	ROLE
Total Clients	Shows overall response volume.
Project-Type Cards	Shows Residential, Commercial, and Industrial totals at a glance.
Project Distribution	Compares the relative share of project types.
Progress Bars	Adds an immediately readable percentage comparison.
Quick Stats	Surfaces the most popular category, response total, and category count.
Export to CSV	Connects dashboard analysis to external data workflows.

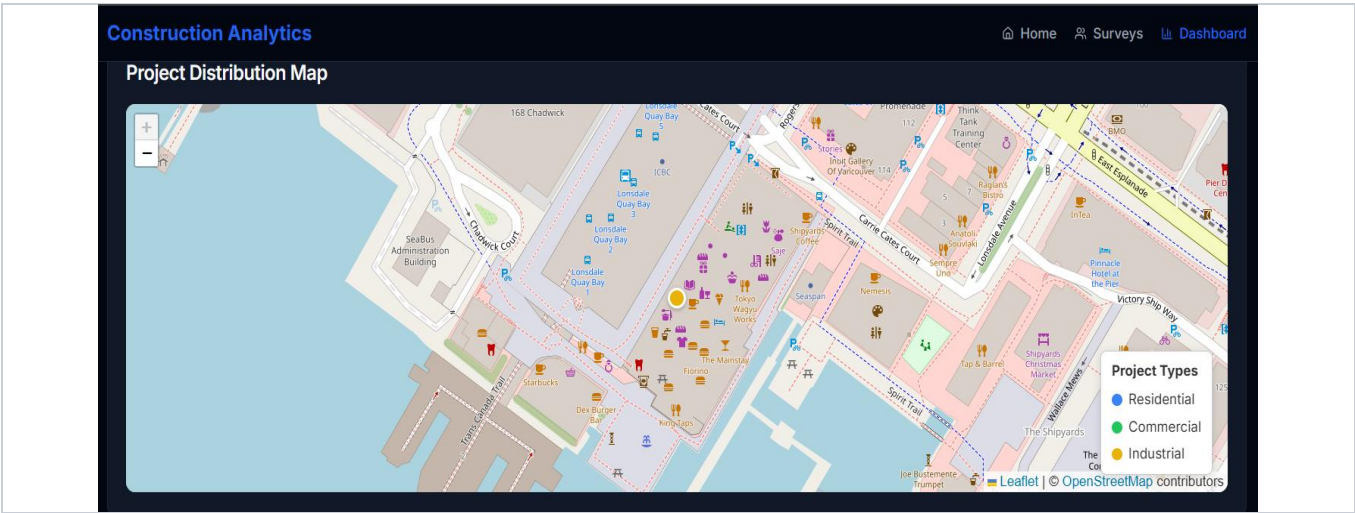
VISIBLE DATA STATE

The supplied screenshot shows 9 clients: 2 Residential, 3 Commercial, and 4 Industrial. Industrial is the largest displayed category at 44.4%.

15 - INTERACTIVE MAP

Project Distribution Map

Leaflet and OpenStreetMap provide the geographic layer. The location is geocoded using Nominatim, and project markers are differentiated by project type.

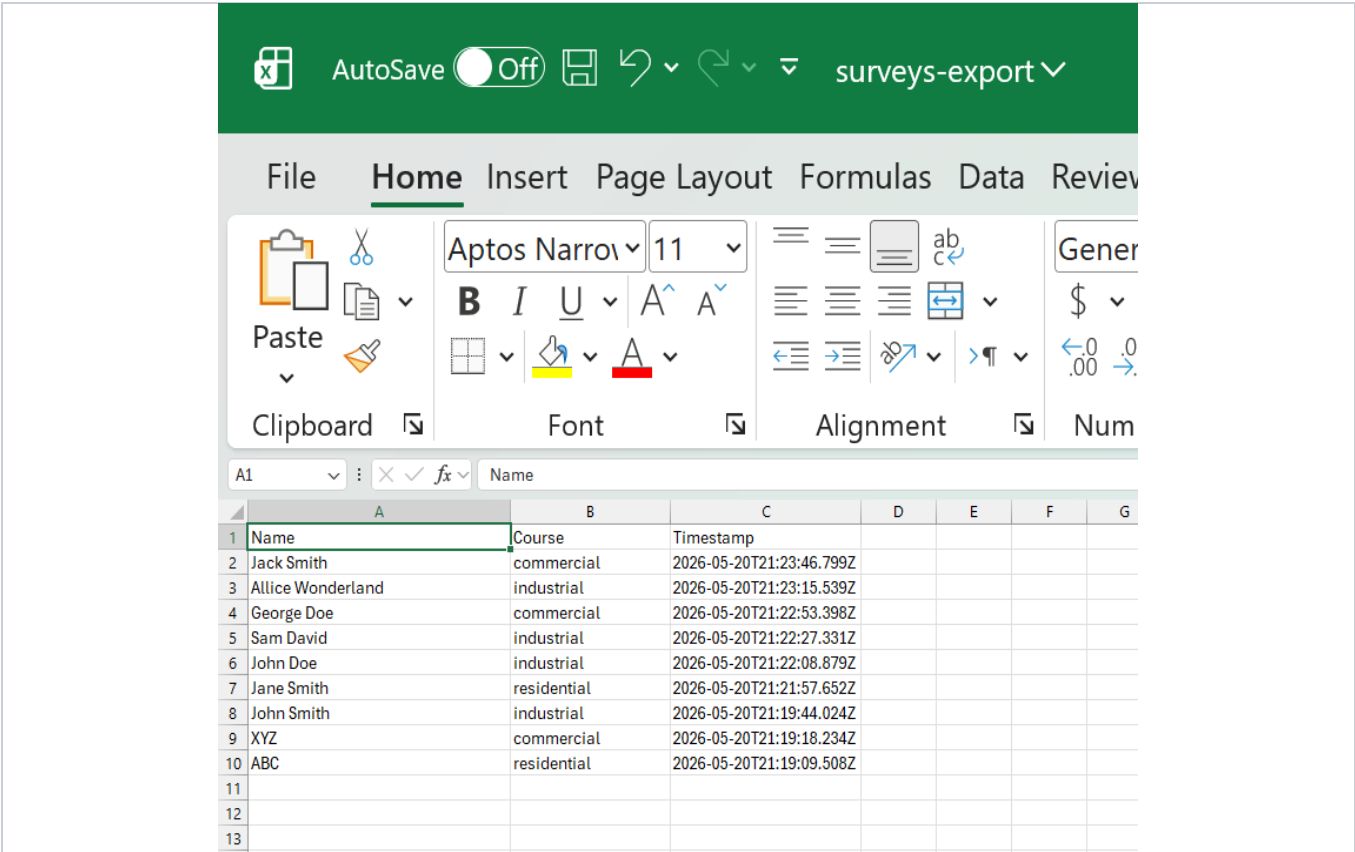


Project Distribution Map (map.png, supplied).

MAP DECISION	WHY
Color-coded markers	Connect geography to the dashboard's project categories.
Popup details	Move from location to record-level information.
Fit all locations	Provide an overview when multiple projects exist.

16 - CSV EXPORT

Export makes the data portable. The documented feature allows users to download survey responses as CSV for Excel, Google Sheets, or other analysis tools.



CSV Export result opened in a spreadsheet (csv.png, supplied).

FIELD	PURPOSE
Client / Company	Identify the response source.
Project Type	Enable category analysis.
Address / Location	Support geographic interpretation.
Timestamp	Support temporal analysis where available.

17 - DESIGN ITERATIONS

The project materials document the final implementation but do not provide a dated sequence of design iterations or user-test feedback. The following is a recommended iteration narrative for portfolio documentation.

01 — Foundation

Problem: Need a clear split between collecting feedback and analyzing it.

Design response: Establish Home → Dashboard structure and primary navigation.

02 — Dashboard Hierarchy

Problem: Multiple metrics compete for attention.

Design response: Prioritize KPI cards, then distribution, then quick stats.

03 — Geospatial Context

Problem: Category counts do not explain where projects are located.

Design response: Add an interactive map and consistent category markers.

04 — External Workflow

Problem: Analytics users may need raw records outside the app.

Design response: Make CSV export a visible dashboard action.

05 — Refinement

Problem: Data surfaces need consistent visual language.

Design response: Align category colours, cards, labels, and interactions.

18 - USABILITY / VALIDATION PLAN

How I Would Validate the Experience

Since the source materials do not include formal usability-test results, this section defines a practical validation plan for the implemented product.

TASK	SUCCESS SIGNAL	WHAT TO LEARN
Submit a response	User completes form without assistance	Form clarity and validation
Identify most popular project type	User answers immediately	Dashboard hierarchy
Find a specific location	User uses map and opens marker	Map discoverability
Explain category colors	User associates marker/chart correctly	Visual consistency
Export dataset	User finds and uses export	CTA discoverability

Interview Questions

- What information do you look for first when reviewing project feedback?
- Would you use the map to make decisions? Why or why not?
- Which dashboard metric is most useful to you?
- What would you expect after clicking Export to CSV?
- Is any part of the dashboard difficult to understand?

19 - FINAL EXPERIENCE

The final product connects four surfaces into one coherent workflow: feedback collection, analytics, geographic exploration, and export.

COLLECT	UNDERSTAND
Validated project information	Cards, charts, percentages

LOCATE	REUSE
Interactive project map	CSV export

END-TO-END EXPERIENCE

The strongest UX story is the continuity from user input to visual insight and reusable data.

20 - TECHNICAL IMPLEMENTATION

Built With

TECHNOLOGY	ROLE
Next.js 14	Application framework and routing
TypeScript	Typed application development
Tailwind CSS	Responsive interface styling
React Query	Data fetching/synchronization
Recharts	Analytics visualization
Leaflet	Interactive map
OpenStreetMap / Nominatim	Map data and geocoding
React Hook Form	Form state and validation
LocalStorage API	Client-side persistence used by the application

Implementation Flow

01	FORM Capture and validate the response.
02	DATA Save and reflect the response in the application.
03	GEOCODE Resolve location into coordinates.
04	VISUALIZE Calculate summaries and render charts.
05	MAP Plot project locations and marker details.
06	EXPORT Generate a portable CSV dataset.

21 - OUTCOME & LEARNINGS

Outcome

Construction Analytics demonstrates an end-to-end product workflow in which a relatively small dataset becomes more useful through visualization, geography, and export.

WHAT THIS PROJECT DEMONSTRATES	EVIDENCE
Data-heavy UI design	Dashboard cards, charts, progress bars, quick stats
Information hierarchy	Overview → distribution → detailed geography
Geospatial UX	Leaflet map + project markers + location details
Design-to-code capability	Next.js + TypeScript + Tailwind implementation
Data workflow thinking	Collection → analysis → export

Key Learnings

- Analytics interfaces benefit from strong hierarchy and progressive disclosure.
- Colour should communicate category meaning consistently across multiple views.
- A map becomes more useful when it is connected to the same data shown in the dashboard.
- Export is part of the user journey, not merely a technical add-on.
- Portfolio case studies should distinguish validated research evidence from design hypotheses.

22 - FUTURE IMPROVEMENTS

Next Opportunities

IMPROVEMENT	VALUE
Filters by project type/date/location	Make larger datasets easier to analyze.
Searchable data table	Connect chart-level patterns to individual records.
Trend analysis	Show how project responses change over time.
Database + authentication	Support persistent multi-user production workflows.
Accessibility audit	Improve keyboard, contrast, form, chart, and map accessibility.
Responsive refinement	Optimize dashboard and map for smaller screens.
Advanced export	Offer filtered exports and additional formats.
Analytics drill-down	Allow category clicks to reveal underlying records.

23 - IMAGE CHECKLIST & PROJECT LINKS

Images to Add & Final Assembly

The four supplied product screenshots are already embedded in this document. The table below tracks status; two optional images remain for a fuller portfolio version.

FILE	STATUS	WHERE	RECOMMENDATION
dashboard.png	✓ Included	Analytics Dashboard / Final Experience	Keep as visual anchor.
home.png	✓ Included	Feedback Form	Full form in normal or success state.
map.png	✓ Included	Interactive Map	Map + one marker popup if possible.
csv.png	✓ Included	CSV Export	Export action + opened CSV if possible.
flow.png	✓ Generated	User Flow (Section 08)	Replace with a Figma export any time if you'd like a different look.
mindmap.png	✓ Generated	Mind Map (Section 09)	Replace with a Figma radial map any time if you'd like a different look.
responsive.png	⚠ Optional	Final Experience	Mobile/tablet view - send this when ready, and I will slot it in.

MINIMUM FINAL IMAGE SET

Every required image is now in the document: dashboard.png, home.png, map.png, and csv.png, plus two generated diagrams (user flow and mind map). The only remaining optional image is a responsive/mobile view — send it whenever you have it, and I'll place it in Section 19.

Project Links

Live Demo: <https://construction-feedback-app.vercel.app/>

Source Code: <https://github.com/mahsakhakpour/construction-feedback>

Portfolio Assets: <https://www.mahsa.website/constructionAnalytics/>